

Quantum and Classical Photonics for Advanced Smart Applications

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Abstract: *We report fiber-based and integrated photonic platforms that use time and frequency control to address scalable quantum communication and classical information processing. Through high-dimensional entanglement, ultrafast time-bin encoding, reservoir computing, spectral machine learning, and microcomb optimization, we demonstrate scalable telecom-compatible architectures for next-generation smart applications. ©2026 The Authors.*

Introduction

Integrated and fiber-based photonic systems are advancing smart technologies for both classical and quantum applications, including information processing, quantum communication, and optical computing [1,2], towards higher speeds, greater parallelism, improved energy efficiency, and enhanced data security. This progress is supported by mature fabrication and control techniques, as well as their inherent compatibility and integrability with existing telecommunications infrastructure. Within this broader landscape, quantum photonics has emerged as a powerful route toward scalable quantum technologies by exploiting the time and frequency degrees of freedom of photons as information carriers [3]. Encoding techniques provide direct access to high-dimensional Hilbert spaces and support robust operations in platforms compatible with telecom-band components. Fiber-based implementations offer high reconfigurability and natural access to temporal processing through delay lines and loop architectures. At the same time, integrated devices provide compactness, phase stability, tunability, and precise control of interferometric circuits and nonlinear quantum light sources.

In parallel, classical photonic systems are driving advances in optical neural networks and related photonic processors for inference and signal transformation, while also enabling learning-assisted control and optimization of photonic devices themselves [4]. This is especially relevant for optical machine learning (ML), edge computing, and intelligent sensing, where conventional

electronic processors face increasing constraints on latency, memory bandwidth, and power dissipation. Here, we present our research progress in fiber-based and integrated photonics for both quantum and classical applications. By controlling light in the time and frequency domains, we develop approaches for quantum-state generation and manipulation in quantum communication and quantum information processing, while also advancing optical neural networks and photonic optimization for next-generation communication, sensing, and high-speed signal processing.

Frequency-bin entanglement

We employ our CMOS-compatible integrated platform, featuring highly-doped silica waveguides that implement low-loss microring resonators (MRRs) and spiral waveguides (SWs), with a strong Kerr response (see Fig. 1(a,b)) [5]. Spontaneous four-wave mixing in MRRs produces broadband photon-pair frequency combs, in which photons are generated in a coherent superposition of multiple discrete frequency modes, yielding high-dimensional frequency-bin-entangled qudit states [6].

Integrated MRR-based devices feature well-defined, discrete frequency modes and passive phase stability, providing a robust, alignment-free source for high-dimensional frequency-bin entanglement. Building on this capability, we further expand the accessible and controllable Hilbert space through hybrid time-frequency approaches. By combining integrated frequency-bin sources with dispersive propagation, electro-optic phase modulation, and time-resolved detection, we coherently control the spectral and temporal degrees

of freedom of photons [7]. These hybrid time-frequency schemes enable measurement-based operations and quantum information processing tasks that are difficult to achieve using purely spectral or temporal encodings and provide a versatile platform for high-dimensional cluster states and other complex entangled photonic resources.

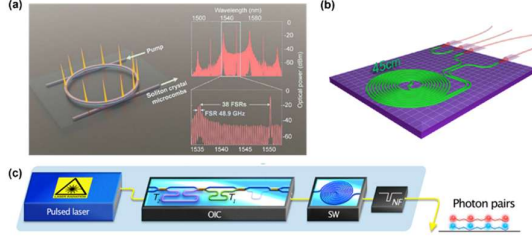


Fig. 1: Integrated devices on our CMOS-compatible platform. (a) Schematic representation of an integrated MRR for the generation of a comb spectrum from four-wave mixing. (b) Schematic representation of an integrated spiral waveguide. (c) Set up for photon state generation via on-chip cascaded interferometers.

Time-bin entanglement

Time-bin encoding provides a complementary and powerful alternative to frequency-bin encoding for generating and manipulating photonic quantum states, enabling distinct implementations and operational strategies based on temporal-mode control. We demonstrate a fiber-based time-bin-encoded photonic quantum walks [8] using two coupled fiber loops of different lengths to establish synthetic temporal photonic lattices, in which controlled delays and dynamic coupling define discrete time bins. Tunable coupling operations act as effective beam splitters between adjacent temporal modes at each time step, implementing discrete-time quantum walks in the temporal domain. Importantly, the system is fully compatible with entangled photon-pair sources, allowing time-bin-entangled states to undergo controlled quantum-walk dynamics. Our results reveal high-visibility two-photon interference over multiple evolution steps (97.82% for $d=2$ and 91.55% for $d=4$), with optimized operating modes that enhance coincidence rates and eliminate post-selection.

Integrated photonics platforms offer ultrafast time-bin control with compactness and enhanced stability [9]. By combining low-loss programmable on-chip interferometric cascades with SWs, we achieve the generation and manipulation of time-bin-entangled qudits with picosecond-scale spacing (see Fig. 1(c)) [9]. This approach, with an experimentally realized time bin spacing of 64 ps (32 ps) for $d=4$ ($d=8$), pushes time-bin encoding into regimes compatible with high repetition rates and dense temporal multiplexing. The high-dimensional time-bin structure maintains compatibility with standard telecom components, enabling the proposed architecture to overcome key rate limitations of conventional time-bin schemes. Such a capability translates directly into practical

applications, including demonstrations of entanglement-based quantum key distribution (QKD) protocols using high-dimensional time-bin states. System operation over 60 km-long fiber links is demonstrated by combining on-chip interferometric stability with high-speed detection.

Dispersion represents a fundamental limitation for time-bin encoding over long distances. We address this challenge using nonlocal strategies that exploit entangled-photon sum-time correlations, resulting in dispersion-resilient two-photon interference [10]. We demonstrate that, by exploiting dispersion cancellation in the summed arrival times of entangled photon pairs, high two-photon interference visibility can be recovered even under extremely large, accumulated dispersion. This effect relies on the nonclassical temporal correlations of entangled photons rather than on dispersion compensation at the receiver, and it is experimentally accessed through post-selection along the sum-correlation (antidiagonal) axis of the joint arrival-time distribution. As a result, we significantly enhance the robustness of time-bin entanglement and improve secret-key rates by more than a factor of 5 in an 80 km-long entanglement-based quantum key distribution link, thereby reinforcing the practical viability of time-encoded quantum communication.

Time-domain optical computing

We achieve time-domain photonic RC with a Sagnac interferometric cavity incorporating an integrated spiral waveguide (SW; Fig. 1(b)) to implement time-delay reservoir computing [11] for optical learning tasks [12]. In this scheme, a temporal input signal excites the cavity, which then generates a high-dimensional dynamical response. The nonlinear reservoir transformation arises from Kerr effects inside the cavity, while the cavity round-trip delay provides the fading memory required for temporal processing. To increase computational throughput, we introduce wavelength-division multiplexing and simultaneously inject several continuous-wave lasers, enabling parallel treatment of independent data streams within a single reservoir. We evaluate the platform on the Mackey-Glass prediction task and obtain a normalized root-mean-square error below 6%. In optical telecom signal recovery, the same system achieves a bit-error rate below 10^{-4} with an effective processing speed of 32.5 GHz.

Frequency-domain photonic processing and optimization

We investigate optical information processing in the spectral domain by exploiting supercontinuum generation in fiber platforms through higher-order soliton fission [13]. Our fully fiber-integrated architecture uses pulses of about 140 fs at 1550 nm with energies of 89 pJ to generate coherent broadband spectra extending from about 1200 to

1750 nm. Computation is performed via a linear readout applied to selected spectral components, thereby avoiding the need to train the nonlinear propagation itself. This framework reaches 100 percent classification accuracy on the IRIS and WINE datasets, a regression error below 0.07 on the Abalone task, and 86.7 percent accuracy on MNIST. In a practical benchmark, speech-based COVID-19 diagnosis reaches 77.1 percent accuracy, outperforming a state-of-the-art digital support vector machine. At the same time, we use learning-driven optimization to control nonlinear spectral states and broadband frequency conversion. In a high-Q integrated microring resonator, shown in Fig. 1(a), we demonstrate ML-assisted control of microcomb states through a genetic algorithm combined with Kerr four-wave mixing [14,15]. A continuous-wave pump excites the resonator, while the output spectrum is tracked as the algorithm iteratively adjusts pump power, detuning, and polarization to optimize spectral flatness and stability. This closed-loop strategy provides reliable access to targeted comb regimes, including stable soliton-crystal states, with coherence up to 86 percent, stability exceeding 24 hours, and repetition rates from 20 to 100 GHz.

Conclusions

Our results show that the degrees of freedom in photon frequency and time provide a common framework for both quantum-state engineering and classical photonic information processing. In the quantum domain, frequency-bin entanglement benefits from integrated nonlinear sources and mature spectral control. At the same time, time-bin encoding and synthetic temporal dimensions enable scalable processing, programmable dynamics, and robustness in fiber networks. Similarly, classical photonic platforms exploit the same temporal and spectral approaches for high-dimensional information processing, optical neural networks, and learning-assisted control of nonlinear photonic states. Fiber-based systems offer reconfigurability, whereas integrated photonics provides stability, compactness, and precise interferometric and spectral control. Hybrid architectures that combine integrated sources with fiber-based temporal and spectral manipulation provide a flexible platform for telecom-compatible quantum communication, signal processing, and photonic ML.

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